

SOS Factorization and Markov–Lukacs Certificates for Real Symmetric Univariate Matrix Polynomials

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1 Overview and Main Theorems

The central result formalized here is a bounded full-line sums-of-squares factorization theorem for real symmetric univariate matrix polynomials. As corollaries, the formalization also gives half-line Markov–Lukacs-type SOS certificates, finite-interval Markov–Lukacs-type SOS certificates, and scalar and constant matrix special cases. The proof uses a Hahn series field with rational exponents as an auxiliary ordered field; the final statements are statements about real polynomial matrices. The full-line theorem follows the mathematical theme of Hanselka–Sinn [1].

Here SOS abbreviates “sum of squares”. In the scalar case this means a finite sum of polynomial squares; in the matrix case it means the corresponding rectangular factorization

$$S = R^T R$$

by a polynomial matrix R .

The Markov–Lukacs-type statements are the half-line and interval forms that express nonnegativity using SOS terms weighted by polynomials nonnegative on the domain. In this project the weights include X , $X - a$, $b - X$, and $(X - a)(b - X)$. For the classical scalar background, see Powers–Reznick [4]. Related work on matrix-polynomial positivity and SOS representations includes Blekherman–Plaumann–Sinn–Vinzant [2] and Klep–Schweighofer [3].

For a real symmetric matrix A , $A \succeq 0$ means $v^T A v \geq 0$ for every column vector v . Thus $P(x) \succeq 0$ for a polynomial matrix P means that the real matrix obtained by evaluating each entry at x is positive semidefinite.

Definition 1 (SOS matrix polynomials and Gram representations). Let $m \in \mathbb{N}$. A matrix polynomial $S \in \text{Mat}_m(\mathbb{R}[X])$ is SOS if there is a polynomial matrix $R \in \text{Mat}_{\ell \times m}(\mathbb{R}[X])$, for some $\ell \in \mathbb{N}$, such that

$$S = R^T R.$$

We call such an R an SOS factor of S .

For $r, m \in \mathbb{N}$, define $Q_{r,m} \in \text{Mat}_{(r+1)m \times m}(\mathbb{R}[X])$ as follows. Its rows are indexed by pairs (k, a) , $0 \leq k \leq r$, $1 \leq a \leq m$, its columns are indexed by $1 \leq b \leq m$, and its entries are

$$(Q_{r,m})_{(k,a),b} = \begin{cases} X^k, & a = b, \\ 0, & a \neq b. \end{cases}$$

For any real matrix $Y \in \text{Mat}_{(r+1)m}(\mathbb{R})$, the product $Q_{r,m}^\top Y Q_{r,m}$ is an $m \times m$ polynomial matrix; if Y is symmetric, then this polynomial matrix is symmetric. A bounded SOS factorization of S with factor degree at most e is equivalent to a Gram representation

$$S = Q_{e,m}^\top Y Q_{e,m}$$

with a real positive semidefinite matrix Y . This is obtained by expanding the entries of the SOS factor in the monomial basis $1, X, \dots, X^e$ and collecting the coefficients into a real matrix. Conversely, the spectral theorem gives $Y = C^\top C$ for a real positive semidefinite Y ; then $R = C Q_{e,m}$ is an SOS factor of degree at most e .

Theorem 2 (Full-line matrix SOS). Let $m, d \in \mathbb{N}$. Let $P \in \text{Mat}_m(\mathbb{R}[X])$ be symmetric, and assume every entry of P has degree at most $2d$. Then the following are equivalent:

- (1) $P(x) \succeq 0$ for every $x \in \mathbb{R}$.
- (2) There is $R \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$, whose entries all have degree at most d , such that

$$P = R^\top R.$$

- (3) There is a real positive semidefinite matrix $Y \in \text{Mat}_{(d+1)m}(\mathbb{R})$ such that

$$P = Q_{d,m}^\top Y Q_{d,m}.$$

Theorem 3 (Half-line Markov–Lukacs certificate). Let $m, d \in \mathbb{N}$, let $M \in \text{Mat}_m(\mathbb{R}[X])$ be symmetric, and let $a \in \mathbb{R}$. Assume every entry of M has degree at most d . Put

$$r(d) = \left\lfloor \frac{d}{2} \right\rfloor, \quad s(d) = \left\lfloor \frac{\max(d-1, 0)}{2} \right\rfloor.$$

Then the following are equivalent:

- (1) $M(x) \succeq 0$ for every $x \geq a$.
- (2) There are $A, B \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$ such that

$$M = A^\top A + (X - a)B^\top B,$$

with entries of A of degree at most $r(d)$, and entries of B of degree at most $s(d)$.

- (3) There are real positive semidefinite matrices

$$Y_1 \in \text{Mat}_{(r(d)+1)m}(\mathbb{R}), \quad Y_2 \in \text{Mat}_{(s(d)+1)m}(\mathbb{R})$$

such that

$$M = Q_{r(d),m}^\top Y_1 Q_{r(d),m} + (X - a) Q_{s(d),m}^\top Y_2 Q_{s(d),m}.$$

Theorem 4 (Finite-interval Markov–Lukacs certificate). Let $m, d \in \mathbb{N}$, let $M \in \text{Mat}_m(\mathbb{R}[X])$ be symmetric, and let $a, b \in \mathbb{R}$ with $a < b$. Put $d_- = \max(d-1, 0)$. If the entries of M have degree at most $2d$, then the following are equivalent:

- (E1) $M(x) \succeq 0$ for every $x \in [a, b]$.

(E2) There are $A, B \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$ such that

$$M = A^\top A + (X - a)(b - X)B^\top B,$$

with entries of A of degree at most d , and entries of B of degree at most d_- .

(E3) There are real positive semidefinite matrices

$$Y_0 \in \text{Mat}_{(d+1)m}(\mathbb{R}), \quad Y_1 \in \text{Mat}_{(d_-+1)m}(\mathbb{R})$$

such that

$$M = Q_{d,m}^\top Y_0 Q_{d,m} + (X - a)(b - X) Q_{d_-,m}^\top Y_1 Q_{d_-,m}.$$

If the entries of M have degree at most $2d + 1$, then the following are equivalent:

(O1) $M(x) \succeq 0$ for every $x \in [a, b]$.

(O2) There are $A, B \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$ such that

$$M = (X - a)A^\top A + (b - X)B^\top B,$$

with entries of A and B of degree at most d .

(O3) There are real positive semidefinite matrices $Y_0, Y_1 \in \text{Mat}_{(d+1)m}(\mathbb{R})$ such that

$$M = (X - a) Q_{d,m}^\top Y_0 Q_{d,m} + (b - X) Q_{d_-,m}^\top Y_1 Q_{d_-,m}.$$

Corollary 5 (Scalar and constant cases). Let $d \in \mathbb{N}$, and let $p \in \mathbb{R}[X]$ have degree at most $2d$. Then p is nonnegative on the real line if and only if $p = q_1^2 + q_2^2$ for some real polynomials q_1, q_2 of degree at most d . Let $m \in \mathbb{N}$, and let $A \in \text{Mat}_m(\mathbb{R})$. Then A is positive semidefinite if and only if $A = R^\top R$ for some $R \in \text{Mat}_{(m+1) \times m}(\mathbb{R})$.

The proof below uses the following rational-function lemma. Its proof is the part that uses the Hahn field, Tsen's theorem, and Pfister forms, and is given in Appendices 付録 C–付録 E.

First define the notation used in the statement. For $r \in \mathbb{R}(X)$, write $\text{NNwd}(r)$ if r is nonnegative at every real point where it is defined: for a representation $r = f/g$ with $g \neq 0$,

$$g(x) \neq 0 \implies \frac{f(x)}{g(x)} \geq 0 \quad (x \in \mathbb{R}).$$

This condition is independent of the chosen representation.

Lemma 6 (Binary sum-of-squares representation). If $a, b \in \mathbb{R}(X)^\times$ satisfy $\text{NNwd}(a)$ and $\text{NNwd}(b)$, then there exist $u, v \in \mathbb{R}(X)$ such that

$$1 = au^2 + bv^2.$$

We first prove the matrix part assuming Lemma 6. The proof of that lemma is given in the appendix.

2 Full-Line Matrix Factorization

Lemma 7. Let $n \in \mathbb{N}$. Let $A \in \text{Mat}_n(\mathbb{R}[X])$ be positive semidefinite at every real point, and suppose $\det A \neq 0$ and $\det A$ is a square in $\mathbb{R}(X)$. Then

$$A = S^\top S$$

for some $S \in \text{Mat}_n(\mathbb{R}(X))$.

Proof. If $n = 0$, the unique empty matrix is the required factor. Assume $n > 0$. Put $F = \mathbb{R}(X)$. Diagonalize the symmetric matrix A over F . Thus there are $C \in \text{GL}_n(F)$ and $d_1, \dots, d_n \in F^\times$ such that

$$C^\top AC = \text{diag}(d_1, \dots, d_n).$$

Positivity at real points shows that each d_i is nonnegative where defined: outside the finite set of poles of C , evaluate the congruence; at any remaining point where d_i is defined, use continuity of d_i and density of the complement of that finite set. Since $\det A$ is a square, write $\det A = r^2$ with $r \in F$. Then

$$d_1 \cdots d_n = \det(C^\top AC) = \det(C)^2 \det(A) = (\det(C)r)^2,$$

so the product of the diagonal coefficients is also a square in F .

The key two-dimensional move is the following. If $a, b \in \mathbb{R}(X)^\times$ are nonnegative where defined, Lemma 6 gives $1 = au^2 + bv^2$. With

$$G = \begin{pmatrix} u & -bv \\ v & au \end{pmatrix} \in \text{GL}_2(F),$$

a direct calculation gives

$$G^\top \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} G = \begin{pmatrix} 1 & 0 \\ 0 & ab \end{pmatrix}.$$

Here $\det G = au^2 + bv^2 = 1$. Thus this congruence transformation extracts a square entry 1 from the two diagonal entries a, b and replaces the remaining information by their product ab . Applying this operation successively gives a matrix $E \in \text{GL}_n(F)$ such that

$$E^\top \text{diag}(d_1, \dots, d_n) E = \text{diag}(1, \dots, 1, d_1 \cdots d_n).$$

Since the last entry is a square, say $d_1 \cdots d_n = (r')^2$ with $r' \in F$, the final diagonal form is $D^\top D$, where

$$D = \text{diag}(1, \dots, 1, r') \in \text{Mat}_n(F).$$

If $C^\top AC = \text{diag}(d_1, \dots, d_n)$ is the initial diagonalization and E is the product of the two-dimensional changes, then $E^\top C^\top ACE = D^\top D$. Hence $A = (D(CE)^{-1})^\top (D(CE)^{-1})$. $\square$

Lemma 8. Suppose $A \in \text{Mat}_n(\mathbb{R}[X])$ and $S \in \text{Mat}_n(\mathbb{R}(X))$ satisfy $A = S^\top S$. Then there are a nonzero polynomial $q \in \mathbb{R}[X]$ and $T \in \text{Mat}_n(\mathbb{R}[X])$ such that

$$q^2 A = T^\top T.$$

Proof. Write $A = S^\top S$ over $\mathbb{R}(X)$. Let q be a common denominator of all entries of S , and put $T = qS$. Then $T^\top T = q^2 A$. $\square$

Lemma 9 (Real-root factor descent). Let $p \in \mathbb{R}[X]$ be irreducible, and suppose $p(\alpha) = 0$ for some $\alpha \in \mathbb{R}$. If $A \in \text{Mat}_n(\mathbb{R}[X])$ and $T \in \text{Mat}_n(\mathbb{R}[X])$ satisfy

$$p^2 A = T^\top T,$$

then there is $U \in \text{Mat}_n(\mathbb{R}[X])$ such that

$$A = U^\top U.$$

Proof. Comparing diagonal entries gives

$$p^2 A_{ii} = \sum_k T_{ki}^2.$$

Evaluating at $X = \alpha$, the left hand side is zero, so

$$\sum_k T_{ki}(\alpha)^2 = 0.$$

This is a sum of squares of real numbers, hence $T_{ki}(\alpha) = 0$ for all k, i . Since an irreducible real polynomial with root α is a nonzero scalar multiple of $X - \alpha$, every T_{ki} is divisible by p . Write $T = pU$, with $U \in \text{Mat}_n(\mathbb{R}[X])$. Substitution gives $p^2 A = p^2 U^\top U$, and cancellation of p^2 gives $A = U^\top U$. $\square$

Lemma 10 (Two-row block construction). Let $q, n \in \mathbb{N}$, and let $N = 2q$ or $N = 2q + 1$. Let $a, b \in \mathbb{R}$ with $b \neq 0$, and put

$$p = (X - a)^2 + b^2 \in \mathbb{R}[X].$$

Let $T \in \text{Mat}_{N \times n}(\mathbb{R}[X])$, and suppose that p divides every entry of $T^\top T$. Then there are a real orthogonal matrix $O \in \text{Mat}_N(\mathbb{R})$ and matrices $H \in \text{Mat}_N(\mathbb{R}[X])$ and $M \in \text{Mat}_{N \times n}(\mathbb{R}[X])$ with the following properties. If $N = 2q$, then

$$OT = HM, \quad H^\top H = pI_N.$$

If $N = 2q + 1$, then

$$OT = HM, \quad H^\top H = \text{diag}(pI_{2q}, p^2).$$

Proof. Put $z = a - ib \in \mathbb{C}$. For a polynomial $f \in \mathbb{R}[X]$, divisibility by p is equivalent to $f(z) = 0$.

From the divisibility assumption, $T(z)^\top T(z) = 0$. Thus the subspace $L \subset \mathbb{C}^N$ generated by the columns of $T(z)$ is totally isotropic for the bilinear form $x^\top y$. Choose a Hermitian orthonormal basis $w_1, \dots, w_r$ of L . Write $w_j = a_j + ib_j$, with $a_j, b_j \in \mathbb{R}^N$. Then

$$\sqrt{2} \Re w_1, \sqrt{2} \Im w_1, \dots, \sqrt{2} \Re w_r, \sqrt{2} \Im w_r$$

is an orthonormal system in $\mathbb{R}^N$. Hence $2r \leq N$, and $r \leq \lfloor N/2 \rfloor = q$. Extend these vectors to an orthonormal basis $e_1, \dots, e_N$. Let $O \in \text{Mat}_N(\mathbb{R})$ be the real orthogonal matrix whose a -th row is $e_a^\top$.

If y is any column of $T(z)$, then $y \in L$, so $y = \sum_j \mu_j w_j$. For $1 \leq j \leq r$, the two coordinates of y measured by e_{2j-1}, e_{2j} are

$$\left(\sqrt{2} a_j^\top y, \sqrt{2} b_j^\top y \right) = \frac{\mu_j}{\sqrt{2}} (1, i).$$

For the remaining basis vectors, the corresponding rows of $OT(z)$ are zero.

Put $B = \begin{pmatrix} X - a & -b \\ b & X - a \end{pmatrix}$. Then $B^\top B = BB^\top = pI_2$. Also $B(z)^\top (1, i)^\top = 0$. For each $1 \leq j \leq q$, let (r_1, r_2) be the two-row pair of OT in rows $2j - 1, 2j$. The matrix

$$B^\top \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

vanishes at z . Since $p \mid f \iff f(z) = 0$, all entries are divisible by p . Define row vectors $m_1, m_2 \in \text{Mat}_{1 \times n}(\mathbb{R}[X])$ by

$$B^\top \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = p \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}.$$

Multiplying by B and using $BB^\top = pI_2$, then cancelling p , gives

$$\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = B \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}.$$

If $N = 2q$, applying this to all q two-row pairs of OT , and placing the obtained m_1, m_2 in the same positions, gives

$$OT = HM, \quad H^\top H = pI_N,$$

with $M \in \text{Mat}_{N \times n}(\mathbb{R}[X])$, where $H \in \text{Mat}_N(\mathbb{R}[X])$ is block diagonal with q copies of B .

If $N = 2q + 1$, the last row of $OT(z)$ is zero, so every entry in that row is divisible by p . Together with the same q two-row blocks, this gives $OT = HM$, where

$$H = \text{diag}(\underbrace{B, \dots, B}_{q \text{ copies}}, p)$$

and $M \in \text{Mat}_{N \times n}(\mathbb{R}[X])$. This H satisfies

$$H^\top H = \text{diag}(pI_{2q}, p^2).$$

□

Lemma 11 (Even-row one-step descent for a no-real-root quadratic). Let $q, n \in \mathbb{N}$. Let $a, b \in \mathbb{R}$ with $b \neq 0$, and put $p = (X - a)^2 + b^2 \in \mathbb{R}[X]$. If $A \in \text{Mat}_n(\mathbb{R}[X])$ and $T \in \text{Mat}_{2q \times n}(\mathbb{R}[X])$ satisfy

$$pA = T^\top T,$$

then there is $U \in \text{Mat}_{2q \times n}(\mathbb{R}[X])$ such that

$$A = U^\top U.$$

Proof. Apply the even-row case of Lemma 10 to T . We get a real orthogonal matrix $O \in \text{Mat}_{2q}(\mathbb{R})$, a polynomial matrix $U \in \text{Mat}_{2q \times n}(\mathbb{R}[X])$, and $H \in \text{Mat}_{2q}(\mathbb{R}[X])$ such that

$$OT = HU, \quad H^\top H = pI_{2q}.$$

Since O is orthogonal,

$$pA = T^\top T = (OT)^\top (OT) = U^\top H^\top H U = pU^\top U.$$

Cancelling p gives $A = U^\top U$. □

Lemma 12 (No-real-root quadratic factor descent). Let $n \in \mathbb{N}$. Let $p \in \mathbb{R}[X]$ be an irreducible quadratic polynomial with no real root. If $A \in \text{Mat}_n(\mathbb{R}[X])$ and $T \in \text{Mat}_n(\mathbb{R}[X])$ satisfy

$$p^2 A = T^\top T,$$

then there is $U \in \text{Mat}_n(\mathbb{R}[X])$ such that

$$A = U^\top U.$$

Proof. Completing the square gives $p = cp_0$, where $c \in \mathbb{R}^\times$ and $p_0 = (X - a)^2 + b^2$, $b \neq 0$. Replacing T by T/c turns $p^2 A = T^\top T$ into $p_0^2 A = (T/c)^\top (T/c)$. Thus it suffices to treat $p = (X - a)^2 + b^2$.

If n is even, write $n = 2q$. Applying Lemma 11 to pA and T gives $pA = M^\top M$ for some $M \in \text{Mat}_n(\mathbb{R}[X])$. Applying the same lemma once more to A and M gives $A = U^\top U$.

Now suppose that n is odd, say $n = 2q + 1$. Apply the odd-row case of Lemma 10 to T . We get a real orthogonal matrix $O \in \text{Mat}_n(\mathbb{R})$, $M \in \text{Mat}_n(\mathbb{R}[X])$, and $H \in \text{Mat}_n(\mathbb{R}[X])$ such that

$$OT = HM, \quad H^\top H = \text{diag}(pI_{2q}, p^2).$$

Write M_{pair} for the first $2q$ rows of M , and M_{rem} for the last row. Using the orthogonality of O , the hypothesis becomes

$$p^2 A = (OT)^\top (OT) = p M_{\text{pair}}^\top M_{\text{pair}} + p^2 M_{\text{rem}}^\top M_{\text{rem}}.$$

Hence

$$p(A - M_{\text{rem}}^\top M_{\text{rem}}) = M_{\text{pair}}^\top M_{\text{pair}}.$$

Apply Lemma 11 to $A - M_{\text{rem}}^\top M_{\text{rem}}$ and M_{pair} . This gives

$$A - M_{\text{rem}}^\top M_{\text{rem}} = U_{\text{pair}}^\top U_{\text{pair}}$$

for some $U_{\text{pair}} \in \text{Mat}_{2q \times n}(\mathbb{R}[X])$. Stacking M_{rem} below U_{pair} gives $U \in \text{Mat}_n(\mathbb{R}[X])$ with $A = U^\top U$. $\square$

Lemma 13 (Denominator descent). Let $n \in \mathbb{N}$. Suppose $A \in \text{Mat}_n(\mathbb{R}[X])$ is positive semidefinite at every real point and

$$q^2 A = T^\top T$$

for some nonzero polynomial q and $T \in \text{Mat}_n(\mathbb{R}[X])$. Then there exists $U \in \text{Mat}_n(\mathbb{R}[X])$ such that

$$A = U^\top U$$

Proof. Choose such a representation with $\deg q$ minimal. We show that this minimal q must be constant. If q is not constant, choose an irreducible factor p and write $q = pq_0$. Rewrite the equality as

$$p^2(q_0^2 A) = T^\top T.$$

If p has a real root, apply Lemma 9 to $q_0^2 A$. If p has no real root, then by the classification of irreducible real polynomials p is quadratic, and we apply Lemma 12 to $q_0^2 A$. In either case we get a polynomial matrix U such that

$$q_0^2 A = U^\top U.$$

Since $\deg q_0 < \deg(pq_0) = \deg q$, this contradicts the minimality of q . Hence q was constant. Since $q \neq 0$, $U = q^{-1}T$ is polynomial and $A = U^\top U$. $\square$

Theorem 2 (Restated; Full-line matrix SOS). Let $m, d \in \mathbb{N}$. Let $P \in \text{Mat}_m(\mathbb{R}[X])$ be symmetric, and assume every entry of P has degree at most $2d$. Then the following are equivalent:

- (1) $P(x) \succeq 0$ for every $x \in \mathbb{R}$.
- (2) There is $R \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$, whose entries all have degree at most d , such that

$$P = R^\top R.$$

(3) There is a real positive semidefinite matrix $Y \in \text{Mat}_{(d+1)m}(\mathbb{R})$ such that

$$P = Q_{d,m}^\top Y Q_{d,m}.$$

Proof. First prove existence of a polynomial factor, without a degree bound, by induction on the matrix size. The case $m = 0$ is given by the empty 1×0 matrix. Assume $m > 0$. If $\det P = 0$, regard P as a linear map $\mathbb{R}[X]^m \rightarrow \mathbb{R}[X]^m$. Since $\mathbb{R}[X]$ is a PID, Smith normal form applies to this map: there are unimodular matrices $C, D \in \text{Mat}_m(\mathbb{R}[X])$ such that

$$DPC = \text{diag}(s_1, \dots, s_{m-1}, 0).$$

Here unimodular means invertible over $\mathbb{R}[X]$, so the inverses also have polynomial entries. The matrix C is the change of basis in the domain, and D is the change of basis in the codomain. Since the last column of the diagonal matrix is zero, if e_m is the last standard basis vector then

$$DPCe_m = 0.$$

Because D is invertible over $\mathbb{R}[X]$, this gives $PCe_m = 0$. Thus the last column $v = Ce_m$ of C lies in the kernel of P .

From this point on we use only the domain change of basis C and transform the quadratic form by congruence:

$$B = C^\top PC.$$

Since PC has zero last column and P is symmetric, the last row and last column of B are zero. Thus

$$B = \begin{pmatrix} B' & 0 \\ 0 & 0 \end{pmatrix},$$

where $B' \in \text{Mat}_{m-1}(\mathbb{R}[X])$ is again positive semidefinite at every real point. By induction,

$$B' = R'^\top R', \quad R' \in \text{Mat}_{m \times (m-1)}(\mathbb{R}[X]).$$

Insert a zero last column to form $\bar{R} \in \text{Mat}_{m \times m}(\mathbb{R}[X])$. Then

$$B = \bar{R}^\top \bar{R}.$$

Therefore

$$P = (C^{-1})^\top B C^{-1} = (\bar{R} C^{-1})^\top (\bar{R} C^{-1}).$$

Finally add one zero row to $\bar{R} C^{-1}$ to obtain the desired $(m+1) \times m$ factor for P .

If $\det P \neq 0$, consider

$$\tilde{P} = \begin{pmatrix} P & 0 \\ 0 & \det P \end{pmatrix}.$$

It is positive semidefinite on the real line and has square determinant:

$$\det \tilde{P} = (\det P)^2.$$

The square-determinant result first gives a factorization $\tilde{P} = S^\top S$ over $\mathbb{R}(X)$. Clearing common denominators gives $q^2 \tilde{P} = T^\top T$ with T polynomial. Applying denominator descent to $\tilde{P}$, we obtain a polynomial matrix $U \in \text{Mat}_{(m+1) \times (m+1)}(\mathbb{R}[X])$ such that

$$\tilde{P} = U^\top U.$$

Split U into its first m columns and its last column:

$$U = (R \mid c), \quad R \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X]), \quad c \in \text{Mat}_{(m+1) \times 1}(\mathbb{R}[X]).$$

Then

$$U^\top U = \begin{pmatrix} R^\top R & R^\top c \\ c^\top R & c^\top c \end{pmatrix}.$$

The upper-left block of $\tilde{P}$ is P , hence $P = R^\top R$.

Finally compare degrees. For each column j ,

$$P_{jj} = \sum_{\alpha} R_{\alpha j}^2.$$

Over the reals, the leading coefficients of nonzero squares are positive, so the top-degree terms in the right hand side cannot cancel. If the right hand side is zero, then every $R_{\alpha j}$ is zero and the bound is trivial. Otherwise

$$\deg P_{jj} = \max_{\alpha} 2 \deg R_{\alpha j}.$$

Since $\deg P_{jj} \leq 2d$, every entry in the j -th column of R has degree at most d .

This proves the bounded SOS factorization from positive semidefiniteness. A bounded SOS factorization gives a Gram representation by expanding the factor R in the monomial basis of Definition 1 and taking the coefficient Gram matrix $C^\top C$. Conversely, if $P = Q_{d,m}^\top Y Q_{d,m}$ with Y positive semidefinite, then for every real x and vector v ,

$$v^\top Q_{d,m}(x)^\top Y Q_{d,m}(x) v = (Q_{d,m}(x) v)^\top Y (Q_{d,m}(x) v) \geq 0.$$

Thus the Gram representation implies positive semidefiniteness on the real line. $\square$

3 Half-Line Corollary

Theorem 3 (Restated; Half-line Markov–Lukacs certificate). Let $m, d \in \mathbb{N}$, let $M \in \text{Mat}_m(\mathbb{R}[X])$ be symmetric, and let $a \in \mathbb{R}$. Assume every entry of M has degree at most d . Put

$$r(d) = \left\lfloor \frac{d}{2} \right\rfloor, \quad s(d) = \left\lfloor \frac{\max(d-1, 0)}{2} \right\rfloor.$$

Then the following are equivalent:

- (1) $M(x) \succeq 0$ for every $x \geq a$.
- (2) There are $A, B \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$ such that

$$M = A^\top A + (X - a) B^\top B,$$

with entries of A of degree at most $r(d)$, and entries of B of degree at most $s(d)$.

- (3) There are real positive semidefinite matrices

$$Y_1 \in \text{Mat}_{(r(d)+1)m}(\mathbb{R}), \quad Y_2 \in \text{Mat}_{(s(d)+1)m}(\mathbb{R})$$

such that

$$M = Q_{r(d),m}^\top Y_1 Q_{r(d),m} + (X - a) Q_{s(d),m}^\top Y_2 Q_{s(d),m}.$$

Proof. After translating $X \mapsto X + a$, it suffices first to treat $a = 0$. If $M(x) \succeq 0$ for all $x \geq 0$, then $P(t) = M(t^2)$ is positive semidefinite for every real t . The full-line theorem gives

$$M(t^2) = R(t)^\top R(t).$$

Here R has $m + 1$ rows. Write each entry of R as an even part plus t times an even part:

$$R(t) = A(t^2) + tB(t^2).$$

Since $M(t^2)$ is even in t , the odd cross term vanishes, and

$$M(X) = A(X)^\top A(X) + XB(X)^\top B(X).$$

The matrices A and B are obtained from the even and odd parts of the same rows of R , so both still have $m + 1$ rows. This proves the existence of the required half-line SOS certificate. Since the entries of $P(t) = M(t^2)$ have degree at most $2d$, the full-line theorem gives $R(t)$ with entry degree at most d . The even-odd decomposition then gives entry-degree bounds $r(d)$ for A and $s(d)$ for B .

The Gram form is obtained from this bounded SOS representation. Expanding the entries of A and B in the monomial bases gives

$$A = CQ_{r(d),m}, \quad B = DQ_{s(d),m}.$$

Therefore $A^\top A$ and $B^\top B$ are represented by the Gram matrices $C^\top C$ and $D^\top D$, which are positive semidefinite.

Returning from the translation gives the general a statement, with weight $X - a$. Conversely, a Gram certificate is positive semidefinite on $x \geq a$ because $Y_1, Y_2 \succeq 0$, sums of positive semidefinite matrices remain positive semidefinite, and the scalar weight $x - a$ is nonnegative on the half-line. Hence positive semidefiniteness, the SOS certificate, and the Gram certificate are equivalent. $\square$

4 Finite-Interval Corollary

Theorem 4 (Restated; Finite-interval Markov–Lukacs certificate). Let $m, d \in \mathbb{N}$, let $M \in \text{Mat}_m(\mathbb{R}[X])$ be symmetric, and let $a, b \in \mathbb{R}$ with $a < b$. Put $d_- = \max(d - 1, 0)$. If the entries of M have degree at most $2d$, then the following are equivalent:

(E1) $M(x) \succeq 0$ for every $x \in [a, b]$.

(E2) There are $A, B \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$ such that

$$M = A^\top A + (X - a)(b - X)B^\top B,$$

with entries of A of degree at most d , and entries of B of degree at most d_- .

(E3) There are real positive semidefinite matrices

$$Y_0 \in \text{Mat}_{(d+1)m}(\mathbb{R}), \quad Y_1 \in \text{Mat}_{(d_-+1)m}(\mathbb{R})$$

such that

$$M = Q_{d,m}^\top Y_0 Q_{d,m} + (X - a)(b - X) Q_{d_-,m}^\top Y_1 Q_{d_-,m}.$$

If the entries of M have degree at most $2d + 1$, then the following are equivalent:

(O1) $M(x) \succeq 0$ for every $x \in [a, b]$.

(O2) There are $A, B \in \text{Mat}_{(m+1) \times m}(\mathbb{R}[X])$ such that

$$M = (X - a)A^\top A + (b - X)B^\top B,$$

with entries of A and B of degree at most d .

(O3) There are real positive semidefinite matrices $Y_0, Y_1 \in \text{Mat}_{(d+1)m}(\mathbb{R})$ such that

$$M = (X - a)Q_{d,m}^\top Y_0 Q_{d,m} + (b - X)Q_{d,m}^\top Y_1 Q_{d,m}.$$

Proof. For a finite interval $[a, b]$, first pull back to the unit interval:

$$N(t) = M(a + (b - a)t).$$

The equivalence between positivity of M on $[a, b]$ and positivity of N on $[0, 1]$ is the affine reindexing theorem.

For N on $[0, 1]$, use the half-line chart

$$t = \frac{u}{1 + u}, \quad u \geq 0.$$

This chart identifies the half-line with $0 \leq t < 1$. The endpoint $t = 1$ is not hit directly, but the identities obtained below are polynomial identities, so equality on $0 \leq t < 1$ extends to the endpoint as well. If the entries of N have degree at most $2d$, then

$$H(u) = (1 + u)^{2d} N\left(\frac{u}{1 + u}\right)$$

is a polynomial matrix of entry degree at most $2d$, and it is positive semidefinite on the half-line. The half-line Markov–Lukacs theorem gives

$$H(u) = A(u)^T A(u) + u B(u)^T B(u).$$

Here the entries of A have degree at most d , and the entries of B have degree at most $\max(d - 1, 0)$.

If $d > 0$, dehomogenizing by $u = t/(1 - t)$ gives the even-bound certificate

$$N(t) = \tilde{A}(t)^T \tilde{A}(t) + t(1 - t) \tilde{B}(t)^T \tilde{B}(t),$$

where

$$\tilde{A}(t) = (1 - t)^d A\left(\frac{t}{1 - t}\right), \quad \tilde{B}(t) = (1 - t)^{d-1} B\left(\frac{t}{1 - t}\right).$$

The degree bounds on A and B make $\tilde{A}$ and $\tilde{B}$ polynomial matrices of degrees at most d and $d - 1$, respectively. If $d = 0$, then $B = 0$ for degree reasons and the same form holds without the second term. The weight $t(1 - t)$ is what becomes $(X - a)(b - X)$ after returning to the original interval. For an odd bound $2d + 1$, the same chart gives

$$H(u) = (1 + u)^{2d+1} N\left(\frac{u}{1 + u}\right),$$

which has entry degree at most $2d+1$ and is positive semidefinite on the half-line. The half-line theorem gives

$$H(u) = A(u)^T A(u) + uB(u)^T B(u).$$

The entries of B have degree at most d . The apparent degree $d+1$ term in A vanishes as follows. For each column j , the diagonal entry satisfies

$$H_{jj} = \sum_{\alpha} A_{\alpha j}^2 + u \sum_{\alpha} B_{\alpha j}^2.$$

The second term has degree at most $2d+1$, and H_{jj} also has degree at most $2d+1$. Thus the coefficient of u^{2d+2} is the sum of the squares of the u^{d+1} -coefficients of the $A_{\alpha j}$, and this sum is zero. Over $\mathbb{R}$, a sum of squares is zero only when every term is zero, so every $A_{\alpha j}$ has degree at most d . Dehomogenization then gives

$$N(t) = t \tilde{B}(t)^T \tilde{B}(t) + (1-t) \tilde{A}(t)^T \tilde{A}(t),$$

with both dehomogenized factors of degree at most d . Finally, the affine inverse $t = (X - a)/(b - a)$ turns this into

$$M(X) = T_0(X) + (X - a)(b - X)T_1(X),$$

in the even case, and

$$M(X) = (X - a)T_0(X) + (b - X)T_1(X)$$

in the odd case, after absorbing the positive scalar factors into the SOS matrices. The Gram versions follow by expanding these bounded SOS factors in the monomial basis and collecting coefficients into positive semidefinite Gram matrices. Conversely, a Gram certificate is positive semidefinite on the interval because the Gram matrices are positive semidefinite and the weights $X - a$, $b - X$, and $(X - a)(b - X)$ are nonnegative on $[a, b]$. Thus, in each parity case, positive semidefiniteness, the SOS certificate, and the Gram certificate are equivalent. $\square$

付録 A Lean Correspondence

The proof above corresponds to the following Lean theorems. Names below are relative to the namespace `MatrixSOS`. The standalone Tsen theorem requires `import MatrixSOS.Proof.Tsen`; the main certificate API is available through `import MatrixSOS`.

- The auxiliary ordered Hahn field is `RationalHahnField`, and its square-root source is `rationalHahnNonnegSquareStatement`.
- The binary sum-of-squares representation is proved as `fracPoly_binaryForm_represents_one_of_nonnegWhereDefined`.
- The common-zero lemma over an algebraically closed field is public as `algebraicallyClosed_commonZero_of_homogeneous`.
- The $K(X)$ case of Tsen's theorem, namely that $K(X)$ is C_1 for algebraically closed K , is public as `Tsen.algebraicallyClosed_ratFunc_c1FieldStatement`.
- The full-line theorem is `fullLine_posSemidef_iff_sos`.

- The bounded right half-line SOS theorem is `Certificates.HalfLine.posSemidefOn_Ici_iff_sos`, and its Gram form is `Certificates.HalfLine.posSemidefOn_Ici_iff_gram`.
- The finite-interval Markov–Lukacs SOS certificates are `Certificates.Interval.posSemidefOn_Icc_iff_lukacsSOS_even` and `Certificates.Interval.posSemidefOn_Icc_iff_lukacsSOS_odd`.
- Their bounded Gram forms are `Certificates.Interval.posSemidefOn_Icc_iff_lukacsGram_even` and `Certificates.Interval.posSemidefOn_Icc_iff_lukacsGram_odd`.
- The scalar and constant special cases are `Certificates.Scalar.polynomial_nonnegOn_univ_iff_sos` and `Certificates.Constant.posSemidef_iff_exists_gram`.

付録 B Preparation for the proof of Lemma 6

The rest of the appendix proves Lemma 6.

Lemma 6 (Restated; Binary sum-of-squares representation). If $a, b \in \mathbb{R}(X)^\times$ satisfy $\text{NNwd}(a)$ and $\text{NNwd}(b)$, then there exist $u, v \in \mathbb{R}(X)$ such that

$$1 = au^2 + bv^2.$$

Here $\text{NNwd}(r)$ means that, for a representation $r = f/g$ with $g \neq 0$,

$$g(x) \neq 0 \implies \frac{f(x)}{g(x)} \geq 0 \quad (x \in \mathbb{R}).$$

The proof of Lemma 6 proceeds as follows. First, the rational-exponent Hahn field is used to detect a real negative value inside a nonnegative-square-closed ordered field. An ordered field K is nonnegative-square-closed if every nonnegative element is a square:

$$0 \leq u \implies \exists v \in K, u = v^2.$$

This negative-value detection shows that a rational function which maps to a nonnegative element in every nonnegative-square-closed ordered extension must satisfy NNwd . Finally, Tsen’s theorem and the splitting criterion for Pfister forms give a local-global principle for ternary quadratic forms, from which Lemma 6 follows.

We use the following quadratic-form terminology in this appendix. Over a field F , the diagonal quadratic form

$$\langle c_1, \dots, c_n \rangle$$

is the form

$$(x_1, \dots, x_n) \mapsto c_1 x_1^2 + \dots + c_n x_n^2.$$

It is isotropic if it has a nonzero zero.

付録 C Rational Hahn Field

We write RHahn for the Hahn series field

$$\mathbb{R}((t^{\mathbb{Q}})) = \mathbb{R}((t^{\mathbb{Q}}))$$

whose series have well-ordered support, equipped with the lexicographic order. For background on Hahn series, see Hahn [8] and Ribenboim [9]. If

$$f = \sum a_q t^q \neq 0,$$

then $\text{ord}(f)$ is the smallest exponent in its support and $\text{lc}(f) = a_{\text{ord}(f)}$. Positivity means $\text{lc}(f) > 0$.

Theorem 14 (Nonnegative elements are squares). If $f \in \text{RHahn}$ and $f \geq 0$, then there exists $g \in \text{RHahn}$ such that $f = g^2$.

Proof. The case $f = 0$ is immediate. Suppose $f > 0$. Let $\alpha = \text{ord}(f)$ and $c = \text{lc}(f) > 0$. Choose $a \in \mathbb{R}$ with $a^2 = c$. Put

$$u = c^{-1} t^{-\alpha} f.$$

Then u has leading term 1, and all terms of $u - 1$ have positive order. Hence the Hahn-series binomial expansion

$$z = \sum_{n \geq 0} \binom{1/2}{n} (u - 1)^n$$

is defined. If $u = 1$, take $z = 1$. Otherwise let $\epsilon = \text{ord}(u - 1) > 0$. The n -th power has order $n\epsilon$, which is unbounded in $\mathbb{Q}$. Thus below any fixed rational bound only finitely many powers contribute, and their supports form a finite union of well-ordered sets. The total support is therefore well ordered, and each exponent receives contributions from only finitely many summands. The usual binomial coefficient calculation then gives $z^2 = u$. Therefore

$$g = a t^{\alpha/2} z$$

satisfies $g^2 = f$. □

付録 D Detecting Negative Values

For each real x , the map $X \mapsto x + t$ sends $\mathbb{R}(X)$ into the Laurent series field around x , and then into RHahn by embedding integer exponents into rational exponents. Nonzero polynomials remain nonzero under this substitution, so their images can be inverted; this defines a unital field embedding and hence the required algebra structure.

Lemma 15. If $p \in \mathbb{R}[X]$ and $p(x) < 0$ for some $x \in \mathbb{R}$, then there is a $\mathbb{R}(X)$ -algebra structure on RHahn for which the image of p is negative.

Proof. Expand p around x :

$$p(x + T) = p(x) + a_1 T + a_2 T^2 + \cdots.$$

The constant term is negative, so this Laurent series is negative in the lexicographic order. The order is preserved by the embedding into rational Hahn series. □

Lemma 16. If $r \in \mathbb{R}(X)$ maps to a nonnegative element in every nonnegative-square-closed ordered $\mathbb{R}(X)$ -algebra, then r is nonnegative at every real point where it is defined.

Proof. If $r = f/g$ were negative at a point x with $g(x) \neq 0$, then $f(x)g(x) < 0$. Lemma 15 applied to the polynomial fg would give a nonnegative-square-closed ordered $\mathbb{R}(X)$ -algebra in which $fg < 0$, and hence $r < 0$, contradicting the hypothesis. $\square$

Lemma 17 (Nonnegative rational functions as sums of two squares). If $r \in \mathbb{R}(X)$ is nonnegative at all real points where it is defined, then

$$r = u^2 + v^2$$

for some $u, v \in \mathbb{R}(X)$.

Proof. Write $r = f/g$, hence $r = fg/g^2$. The polynomial fg is nonnegative on the real line. A nonnegative real polynomial factors into even real roots and positive quadratic factors. The even-root part is a square, and each positive quadratic factor is a sum of two squares. The identity

$$(u^2 + v^2)(s^2 + t^2) = (us - vt)^2 + (ut + vs)^2$$

shows inductively over the factors that fg is a sum of two squares. Dividing by g^2 gives the result. $\square$

付録 E A Ternary Local-Global Step

Let $F = \mathbb{R}(X)$. The proof uses a local-global statement for

$$X^2 = aY^2 + bZ^2.$$

The ordered extensions used here are nonnegative-square-closed ordered fields equipped with an F -algebra structure.

E.1 The $k(X)$ Case of Tsen's Theorem

Definition 18. A field E is C_1 if every homogeneous polynomial over E of degree $d \geq 1$ in more than d variables has a nontrivial zero over E .

The form of Tsen's theorem used here is the rational-function-field case: if k is algebraically closed, then $k(X)$ is C_1 . A broader one-variable function field version is also known as Tsen's theorem, but this proof only uses the rational function field case. The actual application is still narrower: four-variable quadratic forms over $\mathbb{C}(X)$. We use the following common-zero lemma behind the standard proof.

In this subsection, the height $\text{ht}(\mathfrak{p})$ of a prime ideal $\mathfrak{p}$ is the supremum of the lengths n of chains

$$\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}$$

of prime ideals. The height $\text{ht}(J)$ of an ideal J is defined as the minimum height of a minimal prime ideal containing J .^{*1} This is the definition corresponding to Lean's `Ideal.height`.

^{*1} Equivalently, it is the minimum height among all prime ideals containing J . This equivalence follows easily from lemmas available in `mathlib`.

Theorem 19 (Hilbert’s Nullstellensatz). Let k and K be fields, assume that K is algebraically closed, and assume that K is a k -algebra. Put $A = k[Z_1, \dots, Z_M]$. For any ideal $J \subset A$, let $V_K(J) \subset K^M$ be its K -valued common zero locus, and let $I_k(V_K(J))$ be the ideal of k -coefficient polynomials vanishing on that set. Then

$$I_k(V_K(J)) = \sqrt{J}.$$

Proof. In the formalization this is exactly the mathlib theorem

`MvPolynomial.vanishingIdeal_zeroLocus_eq_radical`

□

Lemma 20 (Vanishing ideal of the origin). Let k be a field, and let $A = k[Z_1, \dots, Z_M]$. Then

$$I(\{0\}) = (Z_1, \dots, Z_M).$$

Proof. A polynomial $f \in A$ vanishes at the origin if and only if its constant coefficient is zero. The variable ideal $(Z_1, \dots, Z_M)$ is exactly the set of polynomials with zero constant coefficient. Hence the two ideals are equal. □

Lemma 21 (Radicals do not change height). If A is a commutative ring and $J \subset A$ is an ideal, then

$$\text{ht}(J) = \text{ht}(\sqrt{J}).$$

Proof. This is formalized as `ideal_height_radical`. A prime ideal $\mathfrak{p}$ contains J if and only if it contains $\sqrt{J}$. Therefore the minimal prime ideals containing J and those containing $\sqrt{J}$ are the same, so the minimum of their heights is the same. □

Lemma 22 (Height of the variable ideal). Let k be a field, and let $A = k[Z_1, \dots, Z_M]$. Then

$$\text{ht}(Z_1, \dots, Z_M) = M.$$

Proof. This is formalized as `idealOfVars_height_fintype`. The maximality of the variable ideal is obtained directly from the fact that it is the kernel of the constant-coefficient map $k[Z_1, \dots, Z_M] \rightarrow k$. The empty-variable case is immediate. For the induction step, Lean identifies a multivariable polynomial ring with one more variable as a polynomial ring in one variable over the previous multivariable polynomial ring, using `MvPolynomial.optionEquivLeft`. Heights are transported across this equivalence by `RingEquiv.height_map`, and the height increases by one by `Polynomial.height_eq_height_add_one`. Iterating this over a finite set of variables gives height M . □

Theorem 23 (Krull’s height theorem). Let A be a Noetherian commutative ring, and let $J \subset A$ be a proper ideal. Then

$$\text{ht}(J) \leq \text{spanFinrank}(J).$$

Here `spanFinrank(J)` is mathlib’s finite spanning rank, `Submodule.spanFinrank`. Since ideals in a Noetherian ring are finitely generated, this is the minimum cardinality of a finite generating set of J .

Proof. Lean uses the mathlib theorem `Ideal.height_le_spanFinrank`. □

Lemma 24 (Common-zero lemma). Let k be algebraically closed, and let $P_1, \dots, P_r \in k[Z_1, \dots, Z_M]$ be homogeneous polynomials of positive degree. If $r < M$, then they have a common nontrivial zero in k^M .

Proof. Let $I = (P_1, \dots, P_r)$. Suppose, for contradiction, that the only common zero is the origin.

In Lean, `MvPolynomial.zeroLocus_span` first identifies the zero locus of I with the common zero locus of the generators P_i . Since the P_i are homogeneous of positive degree, the following mathlib lemmas show that the origin is always a zero:

```
MvPolynomial.IsHomogeneous.coeff_eq_zero
MvPolynomial.constantCoeff_eq
MvPolynomial.aeval_zero
```

Applying Theorem 19 to I , and then using Lemma 20, gives

$$\sqrt{I} = (Z_1, \dots, Z_M).$$

By Lemma 21 and Lemma 22, the height of I is M .

On the other hand, Theorem 23 applied to I gives $\text{ht}(I) \leq \text{spanFinrank}(I)$. In the formalization, `Submodule.spanFinrank_span_le_ncard_of_finite` and `Fintype.card_range_le` bound this finite spanning rank by r , because I is generated by the r polynomials P_i . Thus $M \leq r$, contradicting $r < M$. $\square$

Theorem 25 (The $k(X)$ case of Tsen's theorem). If k is algebraically closed, then $k(X)$ is C_1 . In particular, $\mathbb{C}(X)$ is C_1 .

Proof. Let $f(Y_1, \dots, Y_n) \in k(X)[Y_1, \dots, Y_n]$ be homogeneous of degree d , with $n > d$. After clearing denominators, assume that the coefficients of f lie in $k[X]$, and let D be the maximum X -degree of those coefficients. Put $N = D$, and substitute unknown polynomials

$$Y_i(X) = c_{i0} + c_{i1}X + \dots + c_{iN}X^N.$$

After substitution, every coefficient of the resulting polynomial in X is a homogeneous degree- d polynomial in the c_{ij} . There are $n(N+1)$ variables. The number of coefficient equations is at most $D + dN + 1$, because each substituted term has X -degree at most $D + dN$. Since $n > d$, for $N = D$ one has $n(N+1) > D + dN + 1$. Lemma 24 therefore gives a nontrivial choice of the c_{ij} , hence a nontrivial zero of f over $k(X)$. $\square$

E.2 Pfister Forms and the Ternary Local-Global Step

Lemma 26. Let $a, b \in F^\times$. If $X^2 = aY^2 + bZ^2$ has a nontrivial solution over every nonnegative-square-closed ordered field equipped with an F -algebra structure, then it has a nontrivial solution over F .

Proof. Consider the associated two-fold Pfister form

$$\pi = \langle 1, -a, -b, ab \rangle.$$

Over $F(i) \simeq \mathbb{C}(X)$, Theorem 25 gives isotropy of the four-variable quadratic form π . Write a nonzero isotropic vector as $u + iv$, with $u, v \in F^4$. Its real and imaginary parts give $\pi(u) = \pi(v)$ and $B_\pi(u, v) = 0$, where B_π is the associated symmetric bilinear form. If this common value is zero, one of u, v is already a nonzero F -rational isotropic vector. Otherwise the common value is $c \neq 0$, so u, v span a nonsingular plane with diagonal form $\langle c, c \rangle$. Diagonalizing its orthogonal complement gives $\langle c, c, d, e \rangle$. Since $\det \pi = a^2b^2$ and determinants under congruence differ by a nonzero square, $de = s^2$ for some $s \neq 0$. Multiplying the last basis vector by d/s changes its coefficient to d . Thus the form is transformed into

$$\langle c, c, d, d \rangle.$$

The remaining diagonal entries are nonzero, since the Pfister form is nonsingular. The local isotropy hypothesis implies that this diagonal form is isotropic over every nonnegative-square-closed ordered field equipped with an F -algebra structure. For nonnegative-square-closed ordered fields this is equivalent to nonnegativity of $-cd$: in a nonzero zero of $c(u_1^2 + u_2^2) + d(u_3^2 + u_4^2)$, the first sum of squares is nonzero, giving $-c/d = (u_3^2 + u_4^2)/(u_1^2 + u_2^2) \geq 0$. Conversely a square root $r^2 = -cd$ gives the zero $(1, 0, r/d, 0)$. Lemma 16 and Lemma 17 show that $-cd$ is a sum of two squares in F , say $-cd = r^2 + s^2$. Then $(d, 0, r, s)$ is an isotropic vector for $\langle c, c, d, d \rangle$. It is nonzero because $d \neq 0$. Transporting it back gives an isotropic vector for π with coordinates in F . The two-fold Pfister splitting criterion [5] can here be proved directly. Let (x, y, z, w) be such a nonzero zero and put $D = z^2 - aw^2$, so $x^2 - ay^2 = bD$. If $D = 0$, either $(z, w) \neq (0, 0)$ gives $a = (z/w)^2$, or $(x, y) \neq (0, 0)$ gives $a = (x/y)^2$. In either case a is a square and the ternary equation has a nonzero solution with $Z = 0$. If $D \neq 0$, the identity

$$(xz - ayw)^2 - a(xw - yz)^2 = (x^2 - ay^2)(z^2 - aw^2),$$

gives the explicit nontrivial solution

$$(X, Y, Z) = \left(\frac{xz - ayw}{D}, \frac{xw - yz}{D}, 1 \right)$$

of $X^2 = aY^2 + bZ^2$. The ternary form used here is the subform $\langle 1, -a, -b \rangle$; the conventional pure subform of π is instead $\langle -a, -b, ab \rangle$. $\square$

Lemma 27 (Making the first coordinate nonzero). Let E be a field of characteristic 0, and let $a, b \in E^\times$. If $x^2 = ay^2 + bz^2$ has a nontrivial solution over E , then it has one with $x \neq 0$.

Proof. If the original solution has $x_0 \neq 0$, there is nothing to prove. If $x_0 = 0$, then $y_0, z_0 \neq 0$ and $ay_0^2 + bz_0^2 = 0$. Set

$$x = 2ay_0, \quad y = (a + 1)y_0, \quad z = (1 - a)z_0.$$

Then $x \neq 0$, and

$$a((a + 1)y_0)^2 + b((1 - a)z_0)^2 = (2ay_0)^2 + (1 - a)^2(ay_0^2 + bz_0^2) = (2ay_0)^2.$$

$\square$

Lemma 6 (Restated; Binary sum-of-squares representation). If $a, b \in \mathbb{R}(X)^\times$ satisfy $\text{NNwd}(a)$ and $\text{NNwd}(b)$, then there exist $u, v \in \mathbb{R}(X)$ such that

$$1 = au^2 + bv^2.$$

Proof. First show that

$$X^2 = aY^2 + bZ^2$$

has a nontrivial solution over every nonnegative-square-closed ordered F -algebra. By Lemma 17, the functions a and b are sums of two squares in F . In any such ordered field, sums of squares are nonnegative and the images of a, b are nonzero, hence positive. Since nonnegative elements are squares, the image A of a has a square root α , and

$$(X, Y, Z) = (1, \alpha^{-1}, 0)$$

is a nontrivial solution. Lemma 26 gives a nontrivial solution over F , and Lemma 27 lets us choose it with $X \neq 0$. Dividing by X gives $u = Y/X$ and $v = Z/X$, hence

$$1 = au^2 + bv^2.$$

□

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